

(17)
[This question paper contains 8 printed pages]

Your Roll No. :
Sl. No. of Q. Paper : **7743** **I**
Unique Paper Code : 2342202102
Name of the Paper : B. A. (P) NEP
Name of the Course : **Data Interpretation and
Visualization Using
Python**
Semester : II

Time : 3 Hours **Maximum Marks : 90**

Instructions for Candidates :

- (a) Write your Roll No. on the top immediately on receipt of this question paper.
- (b) **Section-A** is compulsory.
- (c) Attempt any **four** questions from **Section-B**.
- (d) Parts of a questions must be answered together.

Assume numpy has been imported as np and pandas has been imported as pd.

Section – A

(Compulsory)

1. (a) What is the purpose of the describe () function in Pandas ? Illustrate with **one** example. 2
- (b) Consider a dataset with a mean of 50 and a standard deviation of 10. Compute the Z – score for a data point 65. 2

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- (c) What are the two advantages of NumPy that makes it important for numerical computation in Python ? 2
- (d) Give the following array : 2
`arr = np.array ([[1, 2, 3], [4, 5, 6], [7, 8, 9]])`
 Write Python code to print the sub - array `[[2, 3], [5, 6]]`
- (e) Differentiate between Experiment and Observational studies. 2
- (f) Give a Series s :
`s = pd.Series ([100, 200, 300], index = [0, 2, 4])`
 Write output of the following Python statements :
- (i) `s1 = s.reindex ([0, 1, 3, 4], method = 'ffill')`
`print (s1)`
- (ii) `s2 = s.reindex ([0, 1, 2, 3, 4], method = 'bfill')`
`print (s2)`
- (g) Write Python code to do the following : 3
- (i) Read a file named "students. csv" into a DataFrame df.
- (ii) Display the first 5 rows of the DataFrame df.
- (iii) Save the DataFrame df to a new file "top_students.csv" using '+' as a separator.
- (h) Write Python command to create the following Series with hierarchical index levels 'Group' and 'Number'. 3

Group	Number	
A	1	10
	2	20
B	1	30
	2	40

- (i) Given the following DataFrame : 3

```
df = pd.DataFrame ({
    'Team' : ['A', 'B', 'A', 'B', 'A', 'B'],
    'Points' : [10, 20, 15, 25, 10, 30]
    'Assists' : [1, 3, 2, 4, 1, 5]
})
```

Write Python code to perform the following operations :

- (i) Group the data by "Team" and compute the total 'Points' per team.
- (ii) Compute the average 'Assists' per team.
- (j) What is the difference between Series and DataFrame in pandas ? Explain with the help of an example of each. 4
- (k) Write Python code to plot a horizontal bar chart using matplotlib for the following data : 4

Subjects = ['Math', 'Science', 'English', 'History']

Scores = [88, 75, 92, 70]

Label the x - axis as 'Scores' and y - axis as 'Subjects' and add a title 'Student Scores by 'Subject'.

Section - B

2. (a) Give an array scores : 4

scores = np.array ([45, 67, 89, 32, 76, 54, 90, 38, 50])

Write Python code for the following :

- (i) Print scores greater than 50.
- (ii) Print scores that are even numbers.
- (iii) Reshape the array scores into a 2-dimensional array scores_2 D of shape 3×3.
- (iv) Convert the data type of scores_2D to np. float 64.

- (b) Given the following DataFrame :

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```
data = pd.DataFrame ({
    'Department' : ['HR', 'IT', 'HR', 'IT', 'HR', 'IT'],
    'Gender' : ['M', 'F', 'M', 'M', 'M', 'F']
    'Salary' : [50000, 60000, 52000, 58000, 51000, 55000]
})
```

Write Python code to perform the following operations :

- (i) Create a pivot table that shows the average salary for each department.
- (ii) Create a pivot table to display average salary by both department and gender.
- (iii) Modify the code in part (ii) to fill missing values in the pivot table with 0.

- (c) Define mean, median, and mode.

6

Give the following dataset :

```
data = [12, 15, 12, 18, 19, 15, 15, 20, 25]
```

Calculate the mean, median, and mode of the data.

3. (a) Differentiate between stack () and unstack () methods with the help of examples.

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- (b) Write Python code to create a 2 D NumPy array of shape 3×3 filled with random integers between 5 and 15 (both inclusive) and then print its dimension, shape and transpose.

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- (c) Write the output of the following Python code :

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```
df1 = pd.DataFrame ({'Math' : 80, 90}, {'English' : [70, 61]},
    index = ['Alice', 'Bob'])
df2 = pd.DataFrame ({'English' : [85, 75], 'Science' : [88, 92]})
```

```

index = ["Bob", "Charile"])
result 1 = df1 + df2
result 2 = df1.add (df2, fill_value = 0)
print ("Result 1 :\n", result1)
print ("\nResult2 : \n", result 2)

```

4. (a) Differentiate between descriptive statistics and inferential statistics with examples. 5

- (b) Given the DataFrame marks :

```

marks = pd.DataFrame ({
    "Math" : [78, 85, np.nan, 88, 74]
    "Science" : [82, np.nan, 95, 91, 76],
    "English" : [75, 80, 88, np.nan, 70]
}, index = ['Amit', 'Rina', 'John', 'Meera', 'Raj'])

```

Write Python code to do the following :

- (i) Compute the average marks for each student (row – wise), ignoring missing values.
- (ii) Identify the name of the student with the highest average marks.
- (iii) Print only those students whose average marks (from part a) are greater than 80.
- (iv) Count the total number of missing values in the DataFrame.
- (v) Fill all missing values in the DataFrame with 'Absent'.

5. (a) Give an array arr :

5

```
arr = np.array ([10, 20, 30, 40, 60, 70])
```


Write Python code for the following :

- (i) Create a slice `s_arr` of `arr` and assign elements from index 2 to 4.
- (ii) Assign a new value (999) to all elements in the `s_arr`.

What will be the contents of `s_arr` and `arr` after the above code is executed ?

(b) Write the output of the following Python code : 10

```
df1 = pd.DataFrame ({
    'Employee ID' : [101, 102, 103, 104],
    'Name' : ['Alice', 'Bob', 'Charlie', 'David'],
    'dept' : ['HR', 'IT', 'Finance', 'Marketing']
})
df2 = pd.DataFrame ({
    'Employee ID' : [102, 103, 105],
    'Salary' : [60000, 75000, 50000],
    'Bonus' : [5000, 7000, 4000]
})
m1 = pd.merge (df1, df2, how = 'inner')
m2 = pd.merge (df1, df2, how = 'left')
m3 = pd.merge (df1, df2, how = 'right')
m4 = pd.merge (df1, df2, how = 'other')
print (m1)
print (m2)
print (m3)
print (m4)
```

6. (a) Give a dictionary data : 4

```
data = {
    'Name' : ['Amit', 'Rina', 'John', 'Meera', 'Raj'],
    'Age' : [21, 23, 25, 22, 24],
    'Score' : [45, 78, 49, 88, 35]
```

Write Python code to perform the following operations :

- (i) Create a DataFrame df from data.
 - (ii) Print all rows of DataFrame df where Age > 22.
 - (iii) Add a new column 'Grade' with values ['A', 'B', 'A', 'C', 'D'] in df.
 - (iv) Delete the column 'Score' from df.
- (b) Explain how missing data is represented in Pandas. With the help of suitable examples, show the use of isnull (), notnull (), dropna (), and fillna () functions in handling missing data. 5
- (c) Write the output of the following Python code : 6

```
I = np.identity (4, dtype = int)
print (I)
diag = I [[0, 1, 2, 3], [0, 1, 2, 3]]
print (diag)
I [[0, 1, 2, 3], [0, 1, 2, 3]] += 5
print (I [[1, 3]])
```

7. (a) Given a Series of exam scores : 5

```
scores = pd.Series ([45, 67, 89, 52, 76, 90, 33, 84,
59, 100])
```

Write Python code to :

- (i) Discretize the scores into bins : 'Poor' (0–50), 'Average' (51–70), 'Good' (71–85) and 'Excellent' (86–100).
 - (ii) Display the frequency count of each bin.
 - (iii) Divide the scores into 4 equal – sized groups (quartiles).
- (b) Write Python code to create a figure with two subplots arranged in a single row using matplotlib.
- 10
- (i) Generate a *blue* coloured line plot of the square of numbers from 1 to 5 in the first subplot. Use a dashed line style ('—') and set the title as 'Line Plot of x^2 '.
Set x-axis label as 'x' and y-axis label as ' x^2 '.
Add custom x-tick labels as ['one', 'two', 'three', 'four', 'five'].
 - (ii) Generate a scatter plot of the same data in the second subplot. Use *green* coloured circle marker ('o') and set the title as 'Scatter plot of x^2 '. Set x-axis label as 'x' and y-axis label as ' x^2 '. Add custom y-tick labels as [1^2 , 2^2 , 3^2 , 4^2 , 5^2].
